Statistical Analysis of Restaurant Tipping Behavior

In this project, we analyze the **Restaurant Tips dataset**. The dataset contains information about restaurant bills, tips, customer characteristics, meal time, day, and party size.

The main objective is to understand:

What factors are associated with restaurant tip amounts, and are the observed relationships statistically significant?

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1. Dataset

The dataset contains the following variables:

Variable	Type	Meaning
total_bill	Quantitative	Total restaurant bill
tip	Quantitative	Tip paid
sex	Categorical	Customer sex
smoker	Categorical	Whether the customer is a smoker
day	Categorical	Day of the week
time	Categorical	Lunch or dinner
size	Quantitative	Number of people in the party

Quantitative variables

Quantitative variables contain numerical measurements.

Examples:

- total_bill
- tip
- size

These variables can be used in calculations such as means, correlations, and regression.

Categorical variables

Categorical variables represent groups or categories.

Examples:

- smoker: Yes/No
- time: Lunch/Dinner
- day: Thur/Fri/Sat/Sun

Categorical variables cannot simply be treated as ordinary numerical measurements. In regression, they are converted into indicator/dummy variables.

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2. Exploratory Data Analysis

What is EDA?

Exploratory Data Analysis (EDA) is the process of examining a dataset before performing formal statistical analysis.

The purpose is to understand:

- Distribution of variables
- Central tendency
- Variability
- Outliers
- Relationships between variables
- Differences between groups

EDA helps us decide which statistical methods are appropriate.

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3. Descriptive Statistics

Descriptive statistics summarize the characteristics of numerical data.

Important measures include:

Mean

The mean is the arithmetic average:

$$\bar{x} = \frac{\sum x_i}{n}$$

For example, the mean tip tells us the average tip paid across the observations.

Median

The median is the middle value when observations are arranged in order.

The median is useful when the data contain extreme values because it is less affected by outliers than the mean.

Standard deviation

Standard deviation measures how spread out observations are around the mean.

A small standard deviation indicates that observations are relatively close to the mean.

A large standard deviation indicates greater variability.

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4. Visualization

Visualization allows us to understand patterns that may not be obvious from numerical summaries.

For this project, we use:

- Histograms
- Box plots
- Scatter plots
- Regression plots
- Q-Q plots
- Residual plots

4.1 Histogram

A histogram shows the distribution of a numerical variable.

For example, a histogram of tip tells us:

- Whether tips are approximately symmetric
- Whether they are skewed
- Whether there are potential extreme observations

This is useful before performing statistical tests and regression.

4.2 Box Plot

A box plot summarizes a numerical variable using:

- Minimum
- First quartile
- Median
- Third quartile
- Maximum

It also helps identify potential outliers.

For example, a box plot of tips by day allows us to visually compare tip distributions across different days.

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5. Correlation

What is correlation?

Correlation measures the strength and direction of association between two quantitative variables.

In this project, we are particularly interested in:

Is there a relationship between total bill and tip?

The Pearson correlation coefficient is:

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Its value lies between:

$$-1 \leq r \leq 1$$

Interpretation

- $r \approx 1$: strong positive relationship
- $r \approx -1$: strong negative relationship
- $r \approx 0$: weak/no linear relationship

If total bill and tip have a positive correlation, it means larger bills tend to be associated with larger tips.

Important limitation

Correlation does **not** prove causation.

A high correlation does not automatically mean one variable causes the other.

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6. Pearson vs Spearman Correlation

We use two correlation measures.

Pearson correlation

Pearson correlation measures **linear association** between two quantitative variables.

It is appropriate when the relationship is approximately linear.

Spearman correlation

Spearman correlation is based on the ranks of observations.

It measures whether two variables have a generally **monotonic relationship**.

It is useful when:

- The relationship is not perfectly linear
- Data contain potential outliers
- Normality assumptions are questionable

Using both provides a more robust view of the relationship between bill amount and tip.

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7. Hypothesis Testing

What is hypothesis testing?

Hypothesis testing is a formal statistical procedure used to determine whether the evidence in a sample is strong enough to reject a proposed null hypothesis.

Every hypothesis test generally contains:

Null hypothesis — H_0

The null hypothesis represents the default assumption.

Alternative hypothesis — H_1

The alternative hypothesis represents the effect or difference we are investigating.

Significance level — α

We use:

$$\alpha = 0.05$$

This means we accept a 5% significance level.

p-value

The p-value tells us how unusual the observed data would be if the null hypothesis were true.

General decision rule:

$$p < 0.05 \Rightarrow \text{Reject } H_0$$

$$p \geq 0.05 \Rightarrow \text{Fail to reject } H_0$$

Importantly, **failing to reject H_0 does not prove that H_0 is true**.

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8. Two-Sample t-Test

Research question

We ask:

Is the average tip significantly different between smokers and non-smokers?

We divide the dataset into two independent groups:

- Smokers
- Non-smokers

We then compare their mean tips.

Hypotheses

Null hypothesis

$$H_0: \mu_{smokers} = \mu_{non-smokers}$$

There is no difference in average tips.

Alternative hypothesis

$$H_1: \mu_{smokers} \neq \mu_{non-smokers}$$

There is a difference in average tips.

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9. Why use a t-test?

A t-test is appropriate when comparing the means of two groups.

Here we have:

Group 1 → Smokers

Group 2 → Non-smokers

and our response variable is numerical:

Tip → numerical

Therefore, a two-sample t-test is suitable.

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10. Why Welch's t-Test?

In our notebook, we use:

```
stats.ttest_ind(
    smoker_tips,
    non_smoker_tips,
    equal_var=False
)
```

The important part is:

equal_var=False

This performs **Welch's two-sample t-test**.

Welch's test does not require the two groups to have identical variances.

This makes it a safer general-purpose choice when group variances may differ.

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11. Interpreting the t-Test

Suppose the resulting p-value is below 0.05.

We would conclude:

There is statistically significant evidence that the mean tip differs between smokers and non-smokers.

If the p-value is greater than 0.05:

There is insufficient statistical evidence to conclude that the mean tip differs between smokers and non-smokers.

The conclusion must be based on the **actual p-value generated by the notebook**.

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12. Effect Size — Cohen's d

Statistical significance does not tell us how large a difference is.

For that reason, we calculate **Cohen's d**.

Cohen's d measures the standardized difference between two group means.

Conceptually:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}}$$

A rough interpretation is:

Cohen's d	Approximate interpretation
0.2	Small
0.5	Medium
0.8	Large

This helps distinguish between:

"There is a statistically significant difference"

and:

"There is a practically meaningful difference."

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13. One-Way ANOVA

Research question

The t-test compares **two groups**.

But day contains more than two groups:

- Thursday
- Friday
- Saturday
- Sunday

Therefore, we need a method capable of comparing multiple group means.

This is where **One-Way ANOVA** is used.

14. ANOVA Hypotheses

Null hypothesis

$$H_0: \mu_{Thu} = \mu_{Fri} = \mu_{Sat} = \mu_{Sun}$$

All group means are equal.

Alternative hypothesis

$$H_1:$$

At least one group mean differs.

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15. Why not perform several t-tests?

Suppose we compare every pair of days using separate t-tests.

That would involve multiple statistical tests.

When many tests are performed, the probability of obtaining a false positive increases.

ANOVA provides an overall test of whether there is evidence that the group means differ.

Therefore:

ANOVA is preferred when comparing more than two group means simultaneously.

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16. Understanding the ANOVA F-Statistic

ANOVA compares two types of variation:

Between-group variation

How different the group means are from each other.

Within-group variation

How much observations vary within each group.

The F-statistic is conceptually:

$$F = \frac{\text{Between-group variation}}{\text{Within-group variation}}$$

A larger F-statistic provides stronger evidence that group means are different.

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17. Interpreting ANOVA

If:

$$p < 0.05$$

we reject the null hypothesis.

We conclude:

At least one day has a statistically different mean tip.

However, ANOVA does **not** tell us which specific days differ.

If ANOVA is significant, a post-hoc test such as Tukey's HSD can be performed to determine which pairs differ.

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18. Kruskal-Wallis Test

We also perform the **Kruskal-Wallis test**.

This is a non-parametric alternative to one-way ANOVA.

Why use it?

ANOVA is a parametric test and relies on assumptions concerning the data and model.

Kruskal-Wallis works with the **ranks** of observations rather than directly relying on their raw values.

This makes it useful when:

- Data are not approximately normally distributed
- Outliers are concerning
- The assumptions of ANOVA are questionable

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19. Parametric vs Non-Parametric Tests

Parametric test

Makes assumptions about the underlying distribution.

Examples:

- t-test
- ANOVA

Non-parametric test

Generally makes fewer distributional assumptions.

Example:

- Kruskal-Wallis

Using both ANOVA and Kruskal-Wallis gives us a useful robustness check.

If both tests lead to similar conclusions, confidence in the overall finding is strengthened.

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20. Ordinary Least Squares Regression

What is regression?

Regression is used to understand the relationship between a dependent variable and one or more explanatory variables.

Our dependent variable is:

tip

Potential predictors include:

total_bill

size

smoker

time

day

The objective is to estimate how these variables are associated with tip amount while controlling for the other predictors.

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21. Simple Linear Regression

A simple linear regression has the form:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

For our problem:

$$tip = \beta_0 + \beta_1 (total\_bill) + \epsilon$$

where:

- Y = tip

- X= total bill
- β_0= intercept
- β_1= slope
- ϵ= error

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22. Multiple Linear Regression

Our final model contains multiple predictors:

$$Tip = \beta_0 + \beta_1 TotalBill + \beta_2 Size + \beta_3 Smoker + \beta_4 Time + \beta_5 Day + \epsilon$$

This is called **multiple linear regression**.

The advantage is that we can examine the relationship between bill and tip while simultaneously accounting for other characteristics.

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23. Why use OLS?

OLS stands for:

Ordinary Least Squares

OLS estimates coefficients by minimizing the sum of squared residuals.

A residual is:

$$e_i = y_i - \hat{y}_i$$

where:

- y_i= actual value
- $\hat{y}_i$= predicted value

OLS chooses coefficients that minimize:

$$\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

In simple terms:

OLS finds the line or plane that gives the smallest overall squared prediction error.

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24. Categorical Variables in Regression

Variables such as smoker, time, and day are categorical.

We use:

C(smoker)

C(time)

C(day)

in the Statsmodels formula.

Statsmodels converts these categories into indicator variables.

For example, smoker might be represented conceptually as:

$$Smoker = \begin{cases} 1 & \text{Yes} \\ 0 & \text{No} \end{cases}$$

The coefficient then represents the expected difference between the two categories, holding the other predictors constant.

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25. Regression Coefficients

Each coefficient describes the estimated relationship between a predictor and the dependent variable.

For example, if:

$$\beta_1 = 0.10$$

for total_bill, then:

Holding the other variables constant, a $1 increase in total bill is associated with an estimated $0.10 increase in tip.

The actual interpretation must use the coefficient produced by the fitted model.

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26. p-Values for Regression Coefficients

Each coefficient has a hypothesis test.

For a coefficient β_j:

Null hypothesis

$$H_0: \beta_j = 0$$

Alternative hypothesis

$$H_1: \beta_j \neq 0$$

If the p-value is below 0.05, we have evidence that the predictor is statistically associated with the response variable after accounting for the other variables in the model.

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27. R-Squared

R^2 measures the proportion of variation in the dependent variable explained by the regression model.

Conceptually:

$$R^2 = 1 - \frac{SS_{Residual}}{SS_{Total}}$$

For example, an R^2 of 0.60 would mean that approximately 60% of the variation in the observed tip amounts is explained by the predictors in the model.

R^2 ranges from 0 to 1 in the standard OLS setting.

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28. Adjusted R-Squared

R^2 generally increases or stays the same when additional predictors are added.

This can be misleading because irrelevant predictors can increase R^2 slightly.

Adjusted R^2 penalizes the model for including additional predictors.

Therefore, it is often more useful when comparing models with different numbers of predictors.

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29. F-Test for Overall Regression

The regression output also includes an F-test.

The null hypothesis is approximately:

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

This means that none of the predictors collectively explains variation in the response.

A significant F-test indicates that the model as a whole provides explanatory information.

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30. Gauss-Markov Assumptions

OLS regression relies on several important assumptions.

The major assumptions we investigate are:

1. Linearity
2. Independence
3. Homoscedasticity
4. Appropriate residual behavior
5. Absence of severe multicollinearity

These assumptions matter because violations can affect coefficient estimates, standard errors, hypothesis tests, and predictions.

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31. Linearity

The relationship between predictors and the expected response should be reasonably linear.

For example, we expect the relationship between:

total_bill → tip

to be reasonably represented by a linear relationship.

We examine this using:

Residual vs Fitted Plot

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32. Residuals

A residual is the difference between an observed and predicted value:

$$e_i = y_i - \hat{y}_i$$

For example:

Actual tip = $5

Predicted tip = $4

Residual:

$$5 - 4 = 1$$

A positive residual means the model underestimated the observation.

A negative residual means the model overestimated it.

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33. Residual vs Fitted Plot

A good residual plot should generally look like a random cloud around zero.

We do **not** want obvious:

- Curves
- Patterns
- Clusters
- Funnel shapes

A curved pattern may indicate that the linear model is missing non-linear structure.

A funnel shape may indicate heteroscedasticity.

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34. Homoscedasticity

Homoscedasticity means:

The variance of the residuals is approximately constant across the range of fitted values.

In simple terms, the model should have roughly similar prediction error variability for small and large predicted values.

If the spread of residuals changes substantially, we have:

Heteroscedasticity

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35. Breusch-Pagan Test

We use the **Breusch-Pagan test** to formally investigate heteroscedasticity.

Null hypothesis

$$H_0\text{:Residual variance is constant}$$

Alternative hypothesis

$$H_1\text{:Residual variance is not constant}$$

If:

$$p < 0.05$$

there is evidence of heteroscedasticity.

If:

$$p \geq 0.05$$

there is insufficient evidence to reject homoscedasticity.

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36. Normality of Residuals

For classical OLS inference, normally distributed residuals are useful, particularly for exact small-sample t and F inference.

We examine residual normality using:

- Histogram
- Q-Q plot
- Shapiro-Wilk test

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37. Q-Q Plot

A Q-Q plot compares the observed residual distribution against a theoretical normal distribution.

If the residuals are approximately normal, the points should lie reasonably close to the diagonal line.

Strong deviations from the line suggest departures from normality.

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38. Shapiro-Wilk Test

The Shapiro-Wilk test formally tests normality.

Null hypothesis

$$H_0\text{:Residuals are normally distributed}$$

Alternative hypothesis

$$H_1\text{:Residuals are not normally distributed}$$

If:

$$p < 0.05$$

we reject the null hypothesis and conclude that the residuals show evidence of non-normality.

However, we should **not rely on the Shapiro-Wilk p-value alone**. We should also examine the Q-Q plot and histogram.

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39. Multicollinearity

Multicollinearity occurs when predictors in a regression model are strongly related to each other.

For example:

total_bill

and

size

may contain some overlapping information because larger parties may tend to generate larger bills.

Severe multicollinearity can make individual coefficients unstable and difficult to interpret.

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40. Variance Inflation Factor — VIF

We use **VIF** to investigate multicollinearity.

Conceptually:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is obtained by regressing predictor j against the other predictors.

Common guidelines are:

VIF	Interpretation
Around 1	Little/no multicollinearity
1–5	Usually acceptable
>5	Potential concern
>10	Often considered serious

These are rules of thumb rather than absolute laws.

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41. Outliers and Influential Observations

An observation can be unusual without necessarily being problematic.

We distinguish between:

Outlier

An observation with an unusually large residual.

High leverage point

An observation with unusual predictor values.

Influential observation

An observation that substantially changes the fitted regression model.

Influential observations deserve investigation because they can strongly affect regression coefficients.

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42. Cook's Distance

Cook's distance measures the influence of individual observations on the fitted regression model.

A commonly used screening threshold is:

$$\frac{4}{n}$$

where $n$is the sample size.

Observations above this threshold should be examined further.

Importantly:

An observation with a large Cook's distance should not automatically be deleted.

We should investigate whether it represents a genuine observation or a data error.
